Karim Taha

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EDUCATION

Cairo University Faculty of Engineering
BSc. Computer Engineering, Grade: Excellent with Honours *Cairo, Egypt*

EXPERIENCE

Backend Engineer (Elixir / OTP, CQRS/ES) **August 2025 - Present**

Sonnen

- Designed and extended CQRS/Event-Sourcing workflows using Commanded: introduced new commands/events and updated projectors and process managers to support cross-service domain logic.
- Resolved a production aggregate state bug by redesigning event emission (two-event sequencing) to correctly handle timeouts and prevent incorrect state replacement involving Oban jobs.
- Re-architected JSON logging pipeline to align with OTP 26+ logger handler,filter,formatter model, adding structured metadata for Phoenix/Plug requests.
- Diagnosed and eliminated shutdown race conditions between dynamic supervisors and PubSub processes.
- Designed and implemented a reusable RabbitMQ client library with explicit producer/consumer behaviours and blocked-connection handling via Phoenix.PubSub state broadcasting.

Parse error recovery and incrementality for GHC **June 2024 - Nov 2024**

Haskell.org (GSoC 2024)

- Contributed to the design and implementation of resumable parsing in the Happy parser generator by specifying the semantics of a ‘catch‘ synchronization token and how it interacts with parser recovery/shift logic to support resumption.
- Extended the `GHC` grammar to accept the ‘catch‘ token at recovery points, enabling parsing to continue and produce partial ASTs instead of failing outright on syntax errors.
- Integrated parser-generator and grammar changes into the compiler pipeline to improve incremental compilation resilience and reduce full-compile failures from parse errors.

Improving FreeBSD User Emulation of the Armv7 Architecture **June 2023 - Sept 2023**

QEMU.org (GSoC 2023)

- Implemented missing FreeBSD syscalls in QEMU’s FreeBSD User (Armv7) to improve user-mode emulation support.
- Handled guest-to-host syscall translation, including parameter marshaling between Armv7 guest and amd64 FreeBSD host address spaces.
- Validated cross-architecture execution by successfully compiling and running C/C++ programs inside an Armv7 jail on amd64 FreeBSD.

PROJECTS

Lang **May 2023**

- A compiler for a statically typed, expression based programming language, written in C.
- The implementation follows the literature standards, by dividing the compiler into a front-end and a back-end.
- The front-end is done using flex/bison parser generators to generate a parser in C.
- The back-end is divided into several stages: symbol tables constuction, type checking and code generation.

RSA Algorithm **June 2022**

- Simple implementation of RSA algorithm in Haskell packaged as a cabal library.
- Used the sockets api for communication between sender and receiver.
- Implemented chosen cipher text attack.

VHDL Pipelined-Processor **Jan 2022**

- A Harvard-architecture 5-stage pipelined processor, implemented in VHDL using Intel’s ModelSim.
- The processor has a RISC-like instruction set architecture, with 8 word-sized general-purpose registers:R0-R7 and some special purpose registers: program counter(PC) And stack pointer(SP).
- The memory address space is 1 MB of 16-bit width and is word addressable. The data bus is 32 bits.

Search engine **June 2021**

- A Search engine in Java, using Java's multi-threading facilities, namely thread pools.
- The Search engine is divided into two parts the web crawler and the indexer.
- The Web Crawler recursively downloads a specified list of web pages, recurring on each sub link.
- The Indexer reads every word, increases it's score then inserts the collected data into an SQL database, ready for query from a simple Web Gui.

TECHNICAL SKILLS

Languages: Elixir, C/C++, Haskell, Python, SQL
Developer Tools: Emacs, Neovim, VS Code, FreeBSD, Linux, Git
Technologies/Frameworks: OTP, CQRS/ES, RabbitMQ, Oban, Phoenix